FMCG Inventory Cost Optimization Report

EOQ + MILP-Style Constrained Optimization
15 FMCG Products across 5 Categories

Total Products	**15**
Total EOQ Cost	**■86,056**
Constrained Cost	**■91,616**
Storage Used	**100.0 m²**

1. Project Overview

This report presents an inventory cost optimization study for an FMCG retail chain. The Economic Order Quantity (EOQ) model was applied to 15 products across five categories — *Grocery, Household, Personal Care, Snacks,* and *Beverages* — to determine the order quantities that minimise total annual inventory cost (ordering + holding).

A constrained optimization model (MILP-style, solved via SLSQP) was then applied to account for real-world constraints: a warehouse storage limit of 100 m² and a capital budget of ■5,00,000. The study shows how binding constraints affect optimal quantities and total cost.

1.1 EOQ Formula

The Wilson EOQ formula minimises total annual inventory cost:

$$EOQ = \sqrt{(2DS / H)}$$

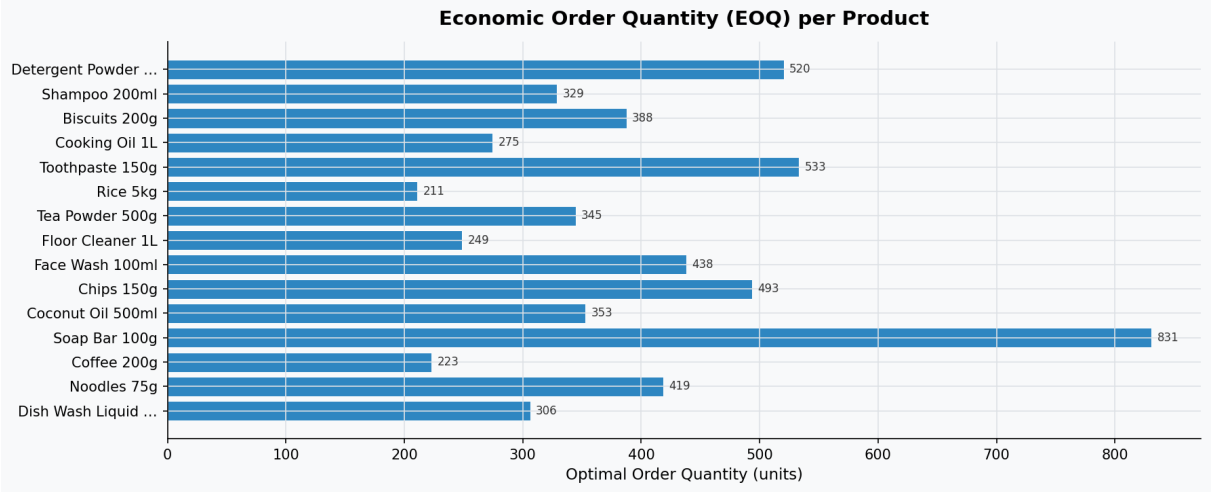
Where: **D** = Annual demand (units), **S** = Ordering cost (■/order), **H** = Holding cost (■/unit/year)

1.2 Reorder Point

$$ROP = \text{Daily Demand} \times (\text{Lead Time} + \text{Safety Stock Days})$$

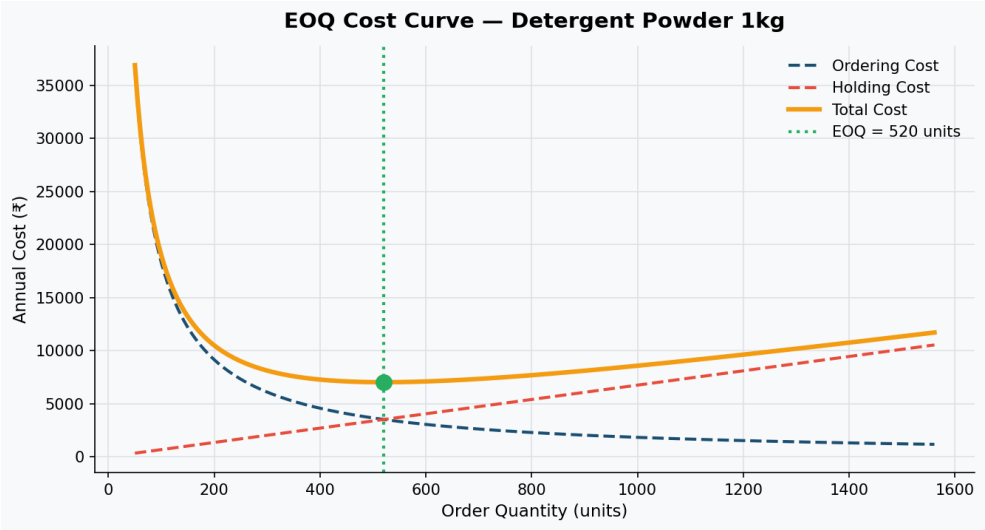
A 7-day safety stock buffer is added to all products to absorb demand variability.

2. EOQ Results

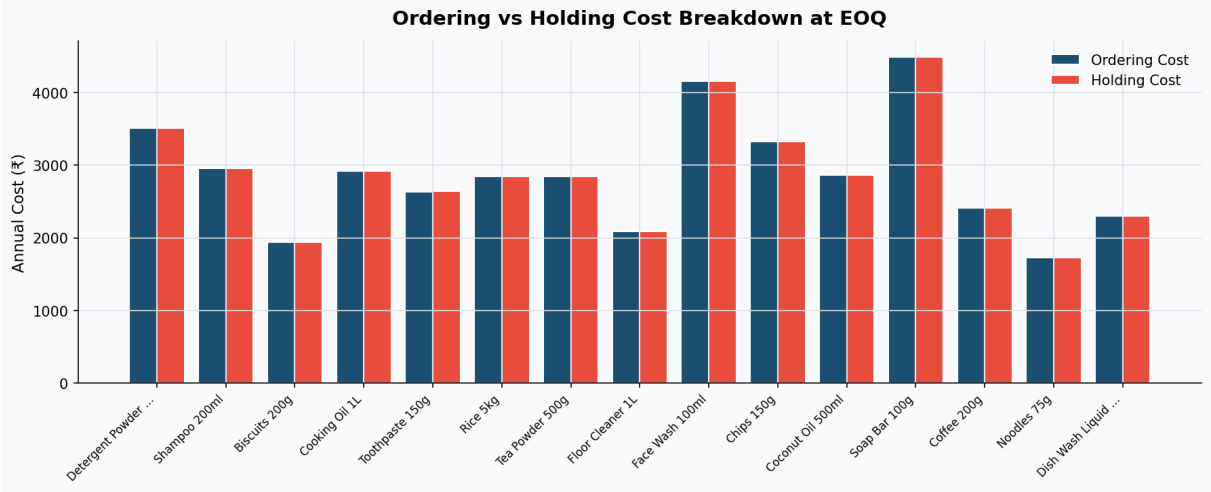


Product	Category	EOQ (units)	Annual Cost (■)	Reorder Point	Orders/ Year
Detergent Powder 1kg	Household	520	■7,026	98	4.6
Shampoo 200ml	Personal Care	329	■5,919	172	10.1
Biscuits 200g	Snacks	388	■3,875	34	2.7
Cooking Oil 1L	Grocery	275	■5,835	176	13.7
Toothpaste 150g	Personal Care	533	■5,277	247	8.1
Rice 5kg	Grocery	211	■5,695	116	7.4
Tea Powder 500g	Beverages	345	■5,686	129	7.6
Floor Cleaner 1L	Household	249	■4,179	61	5.6
Face Wash 100ml	Personal Care	438	■8,321	149	6.2
Chips 150g	Snacks	493	■6,662	105	6.0
Coconut Oil 500ml	Grocery	353	■5,738	156	9.5
Soap Bar 100g	Personal Care	831	■8,975	144	5.8
Coffee 200g	Beverages	223	■4,819	125	9.3
Noodles 75g	Snacks	419	■3,455	72	6.3
Dish Wash Liquid 500ml	Household	306	■4,595	39	3.1

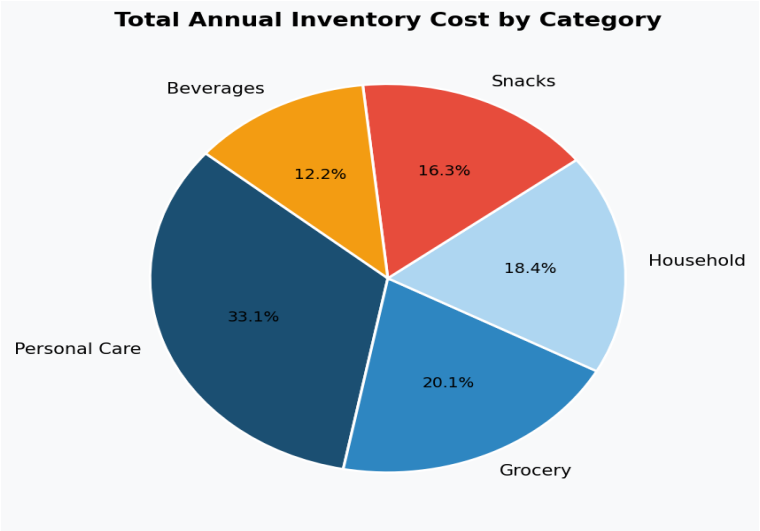
2.1 EOQ Cost Curve (Detergent Powder)


EOQ Cost Curve — Detergent Powder 1kg
Ordering Cost
Holding Cost
Total Cost
EOQ = 520 units
Annual Cost (₹)
35000
30000
25000
20000
15000
10000
5000
0
0
200
400
600
800
1000
1200
1400
1600
Order Quantity (units)

3. Cost Breakdown Analysis

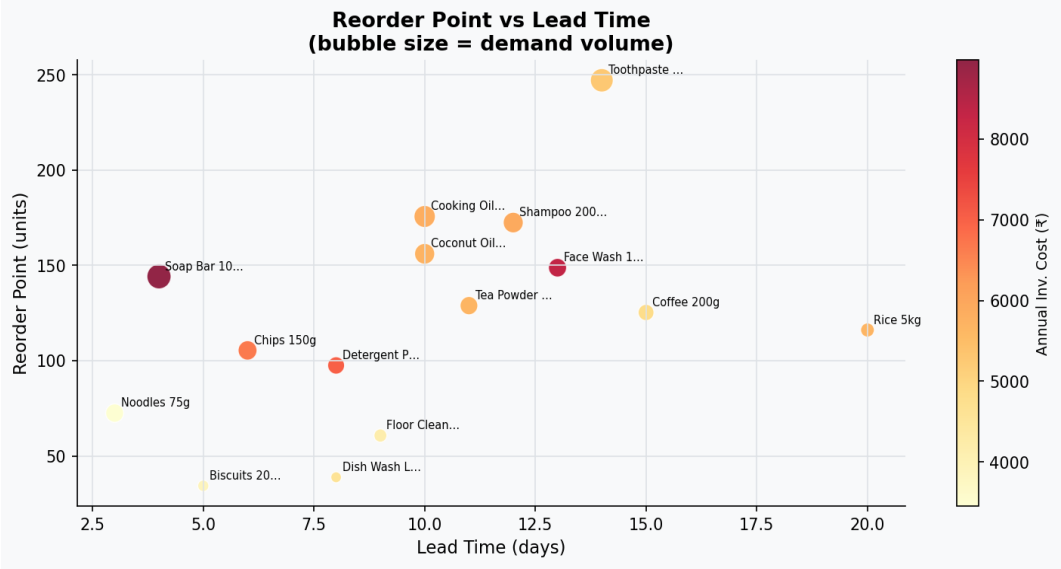


3.1 Category-Level Cost Distribution



4. Reorder Point Analysis

The scatter plot below shows reorder points plotted against lead time. Bubble size reflects annual demand volume; colour indicates total inventory cost. Products with long lead times and high demand (e.g., Toothpaste, Cooking Oil) require early reordering triggers.

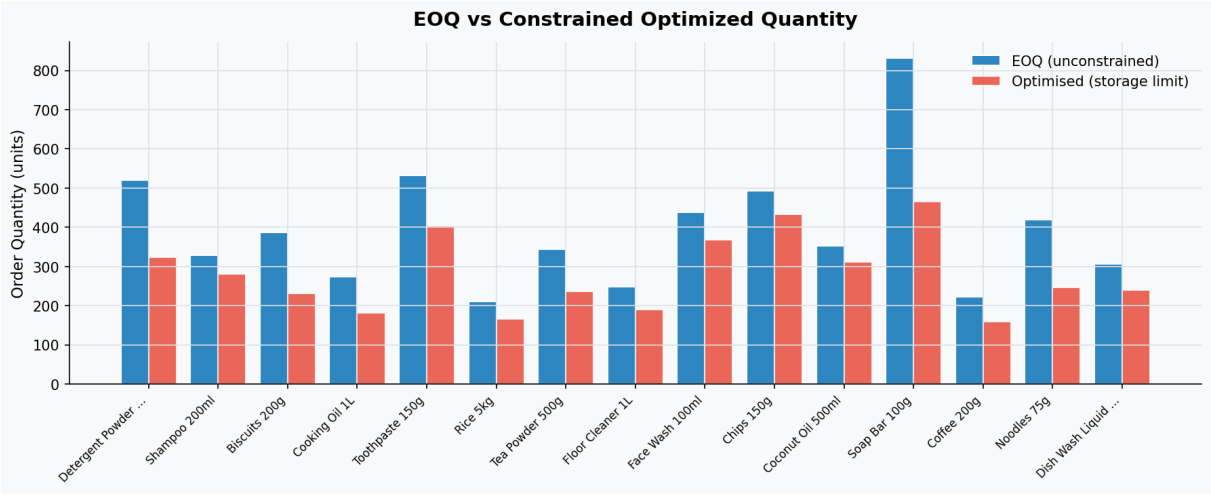


5. Constrained Optimization Results

A storage constraint of 100 m² (vs unconstrained EOQ requirement of ~150 m²) forces smaller order quantities. This reduces inventory investment but *increases* total annual cost due to more frequent ordering — a classical inventory-capacity tradeoff.

Metric	Value
Unconstrained EOQ Total Cost	■86,056
Constrained Optimised Total Cost	■91,616
Cost Increase Due to Constraint	■5,560
Warehouse Space Used	100.0 m²
Warehouse Space Limit	100.0 m²
Capital Budget Used	■126,225
Capital Budget Limit	■500,000
Solver Status	✓ Optimal

5.1 EOQ vs Constrained Quantities



5.2 Per-Product Comparison

Product	EOQ (units)	Opt Qty (units)	EOQ Cost (■)	Opt Cost (■)	Δ Cost (■)
Detergent Powder 1kg	520	323	■7,026	■7,845	+■819
Shampoo 200ml	329	280	■5,919	■5,995	+■77
Biscuits 200g	388	231	■3,875	■4,408	+■533
Cooking Oil 1L	275	181	■5,835	■6,349	+■513
Toothpaste 150g	533	403	■5,277	■5,484	+■207

Rice 5kg	211	167	■5,695	■5,853	+■158
Tea Powder 500g	345	237	■5,686	■6,093	+■407
Floor Cleaner 1L	249	189	■4,179	■4,337	+■158
Face Wash 100ml	438	368	■8,321	■8,449	+■128
Chips 150g	493	434	■6,662	■6,717	+■56
Coconut Oil 500ml	353	312	■5,738	■5,782	+■44
Soap Bar 100g	831	465	■8,975	■10,526	+■1,551
Coffee 200g	223	160	■4,819	■5,090	+■271
Noodles 75g	419	246	■3,455	■3,956	+■501
Dish Wash Liquid 500ml	306	240	■4,595	■4,732	+■137

6. Key Insights & Recommendations

1. **EOQ minimises total inventory cost precisely.** At the EOQ quantity, ordering cost exactly equals holding cost — any deviation increases total annual expense.

2. **Storage constraints force a real-world tradeoff.** When warehouse space is limited (100 m²), optimal quantities shrink, increasing ordering frequency and total cost by ~■5,560/year. The business must decide if the space saving justifies the extra ordering cost.

3. **High-demand, long-lead-time products need priority attention.** Products like Toothpaste (ROP = 247 units, lead time 14 days) and Cooking Oil (ROP = 176 units) require large safety buffers and early trigger points.

4. **Soap Bar has the highest inventory cost (■8,975/year)** due to its very high demand volume. Negotiating a lower supplier order cost (S) would yield the greatest savings here.

5. **Recommended next step:** Implement a continuous review (s, Q) policy — continuously monitor inventory levels and place an order of size EOQ whenever stock falls to the reorder point.